Problem Chosen	2023 MCM/ICM Summary Sheet	Team Control Number
C		2310518

Unveiling the Mystery of Wordle

Summary

After four days of ample modeling and data analysis, we have made a series of achievements and interesting discoveries around Wordle.

For problem 1, involving the explanation of the variability observed in the reported results and the prediction of its value by March 1, 2023, a joint prediction model was developed. To explicate the sudden surge in the time series, the **SEIRS model in epidemiology** was employed, as it provides a robust interpretative framework and aligns closely with the temporal trends observed in the data. The residuals generated by the model exhibit stability, and these were analyzed using the **classical ARIMA model**, which enables the characterization of residual properties. Furthermore, word features were extracted from given data.After **correlation analysis** with the percentage of scores reported that were played in Hard Mode,the results obtained indicate that these factors exhibit no association with the percentage of scores reported.

For problem 2, we leveraged several word attributes obtained from publicly available databases, such as the **COCA corpus** and the **NLTK library**. Specifically, we extracted features such as word frequency and lexicality, which were used as inputs for **decision tree regression** to predict the distribution of "tries". Our model was optimized to enhance its predictive ability, despite the limited sample size, and was found to outperform other commonly used models such as neural networks and SVR, based on performance metrics evaluated on the validation set. We further applied our optimized model to predict the "tries" distribution for the word "errie" and are confident that **our approach yields more accurate predictions compared to other models**.

For problem 3, we used the average number of attempts as input to cluster using the k-means algorithm, and since the variables did not reuse the word features, we only used the determined number of categories, fitting the **Gaussian mixture model** with the previously mentioned word features as input, and ended up with 3 classes. After analysis, we determined that these 3 classes were classified according to "difficulty". Finally we used the model to classify "eerie" and the results showed it to be of medium difficulty

Finally, we also found some interesting discoveries from the dataset (most of which have been analyzed in the first half of the paper, so we only list new findings). For example, as time passes and novelty wears off, the public's attention gradually shifts away from the popular Wordle game. Although Wordle's popularity is not as high as before, the majority of those who have lost interest are not core players.

Keywords: **SEIRS model; ARIMA; NLP; decision tree model; K-means; Gaussian mixture model;**

Contents

1 Data preprocessing **2**

2 SEIRS-ARIMAM model **3**

2.1 SEIRS model 3

2.2 ARIMA model 4

2.3 Combined SEIRS-ARIMA model 7

3 The attributes affects the percentage **7**

4 Prediction of "tries" of words **9**

4.1 Feature engineering 9

4.2 Comparison of multiple regression models 11

4.3 Hyperparameters of Tree 13

4.4 Prediction of "errie" 15

5 Difficulty differentiation model for words **16**

5.1 K-means to determine the number of classifications 16

5.2 Gaussian mixture model 16

6 Other features in the data **18**

6.1 Feature 1 18

6.2 Feature 2 19

Letter **20**

A Search for the best value of Decision Tree Depth **22**

1 Data preprocessing

To better solve the problem, we obtained 60,000 words commonly used in the coca corpus (4259 words of length 5). We extracted the usage frequency of the words from them (which contains speaking frequency, literature frequency, magazine frequency, news usage frequency and academic frequency considering that not most people know academics, we did not include the academic frequency of the words used).

In addition,initial data processing is required to rectify inaccuracies within the raw dataset. After importing the data with the aid of the pandas library in Python, an error was reported upon calling the data, leading to the discovery of erroneous English words within the dataset. As such, corrective measures are necessary for ensuring the accuracy of the data.We cleaned the data as the following steps(You can also see the summary in the 1):

step 1. There are some words that do not appear in the corpus.we set the usage frequency to 10(relatively small).

step 2. Some words have missing letters or have extra letters, for example 'rprobe' for 22/12/16 should be 'probe' and 'tash' for 22/4/29 should be 'trash'.We replace theme by right words

step 3. Some words are not English letters, for example 'naïve' for 22/12/11 should be 'naive'.We use i to replace it.

step 4. There are also words with unknown space characters at the beginning or end, which is detrimental to the data analysis of the string, e.g. 'favor ' for 22/1/12 should be 'favor'.

step 5. In the "Number of reported results" column, the data of 2022/11/30 is 2569, however, both its upper and lower data are more than 20,000.We use the average of the top and bottom two days for substitution

step 6. In the "Number in hard mode In the "Number in hard mode" column, the data of 2022/2/13 is 3249, however, its upper and lower data are close to 10,000. We use the average of the top and bottom two days for substitution

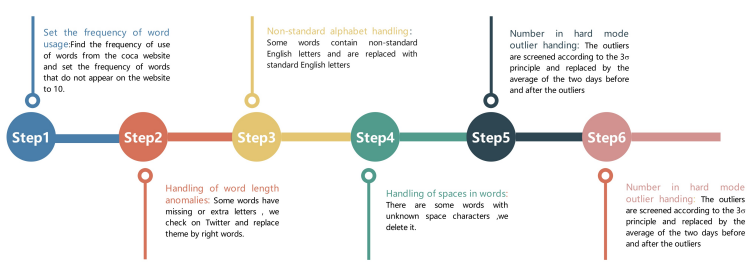


Figure 1: The step of data preprocessing

2 SEIRS-ARIMAM model

2.1 SEIRS model

The fundamental of SEIRS model

The SEIR(Susceptible-Exposed-Infected-Recoverd) model,a well-established and robust traditional infectious disease model[4], is a non-linear kinetic model that enables a comprehensive study of the transmission rate, the number of infections, and the temporal dynamics of infectious diseases[4]. A key feature of the SEIR model is its incorporation of the exposed group concept, and its division of the population into four categories - susceptible, exposed, infected, and recovered. The model further establishes differential equations through the inter-relationships between these four categories, providing valuable guidance for the development of effective strategies for the control and prevention of infectious diseases.

The SEIR model categorizes individuals based on their infection status. The **susceptible group (denoted as S)**comprises individuals who have not yet contracted the disease and remain healthy, constituting a large proportion of the population. The **exposed group(denoted as E)** encompasses individuals who are in the incubation period of the disease, while the **infected group(denoted as I)** consists of those who have received a diagnosis of the disease. The **recovered group(denoted as R)** includes individuals who have been removed from the infected group, and comprises those who have either recovered or have passed away.And the SEIR model employs several parameters to capture the infection status of individuals within each group.

Denote the transmission coefficient by $\boldsymbol{\beta}$,which is a crucial parameter that represents the **average number of infectious contacts** made by infected individuals within the population during a given time period. Higher values of β correspond to greater opportunities for the virus to spread.The correlation between β, S, and I can be depicted by the following equation:

$$\frac{dS}{dt} = -\frac{\beta SI}{N} + \xi R$$

where t represents time, and N represents the population size, which is the sum of individuals in the susceptible, exposed, infectious, and recovered groups. Additionally, the **rate of loss of immunity among the recovered individuals is denoted by $\boldsymbol{\xi}$**(The detail of it will be discussed below).Furthermore,denote $\boldsymbol{\sigma}$ as the rate at which exposed individuals transition to the infectious phase. This rate is quantified as **the reciprocal of the average duration of the incubation period** and reflects the speed at which the incubation period of the disease concludes and the individual becomes infectious. Individuals can leave E by entering the infected group, as determined by the following equation:

$$\frac{dE}{dt} = \frac{\beta SI}{N} - \sigma E$$

Once an individual is infected, there is a period of time before they move to the recovery group, and the rate at which this occurs is denoted as γ. In other words, γ **represents**

the number of individuals who recover from the infection per unit time.The following equation depicte the correlation between γ,I and E:

$$\frac{dI}{dt} = \sigma E - \gamma I$$

Upon recovery, an individual may shift from the recovery group to the susceptible group due to loss of immunity, as characterized by the rate parameter ξ.We have the equation:

$$\frac{dR}{dt} = \gamma I - \xi R$$

The entire model framework is shown in the Figure 2

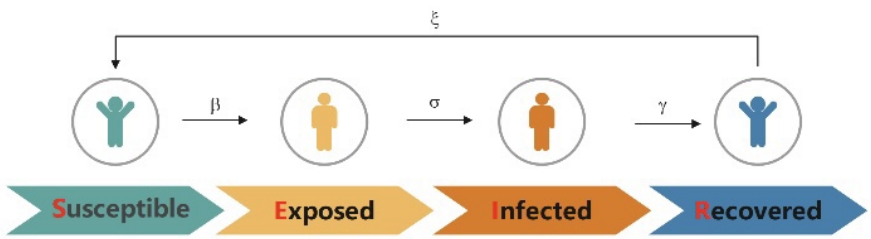


Figure 2: The distribution of each group

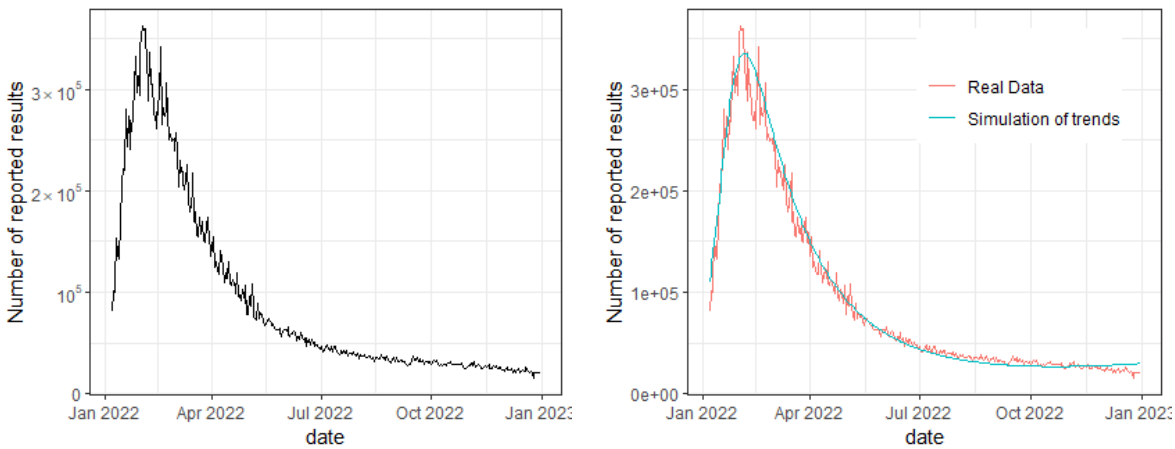
The result of SEIRS model

In this section, the SEIRS model parameters are established to attain a satisfactory fit with the data while also offering a cogent explication of the parameters' implications within the context of our problem.

The dataset reveals that the origin of the game under investigation did not occur on the earliest day represented in the data. Additionally, explosive growth, as depicted in Figure 3(a), did not commence on the earliest day represented in the data. Therefore, it is necessary to adjust the SEIRS model's starting point to a date 40 days prior to 2022-1-7(2021-11-28) to ensure accurate simulation of the potential pattern.On 2021-11-28, the values assigned to the SEIR model parameters are as follows: S = 550,000, E = 15,000, I = 10,000, and R = 200. These quantities correspond to the number of individuals who are susceptible to become potential players of the wordle game and those who have already been exposed, the number of individuals who are currently infected with the game, and those who have already recovered, respectively. Specifically, the number of individuals who are attracted to playing the wordle game is 15,000, while the number of individuals who have already initiated the gameplay is 10,000. Furthermore, the number of individuals who have played the wordle game and abandoned it is 200.Figure 3(b) illustrates the curve that has been fitted to the data, displaying a strong adherence to the prevailing trend of the dataset.

2.2 ARIMA model

Utilizing the SEIRS model, the time series was approximated and interpreted effectively.Then we get the residual y_t by the equation $y_t = x_t - I$. To enhance the model's



(a) tendency (b) simulation

Figure 3: tendency and the simulation of it

precision, an ARIMA(Auto-Regressive Integer Moving Average) model was employed to scrutinize the characteristics of the residuals.The comprehensive model framework is illustrated in Figure 3.

The fundamental of ARIMA model

Time series forecasting frequently utilizes the ARIMA model as an approach that models the data series of the forecast object as a stochastic process and utilizes a specific mathematical model to approximate the underlying series[5]. The incorporation of residuals obtained from the discrepancy between actual values and the simulated trend of the SEIRS model into the ARIMA model enables the forecasting of future residuals. These predicted residuals are then combined with the SEIRS trend to produce a robust forecast of the actual values[8]. Key parameters of the ARIMA(p, d, q) model include p, which denotes the number of lags of the time-series data itself used in the prediction model, d, which represents the number of orders of differencing required to stabilize the time-series data, and q, which signifies the number of lags of the prediction error used in the prediction model.The relationship between them can be characterized by the subsequent equation.

$$(1-\phi_1 B-\cdots-\phi_p B^p)(1-B)^d y_t \quad = c+(1+\theta_1 B+\cdots+\theta_q B^q)\,\varepsilon_t$$

where B is the lag operator,y_t is the time series,c is a constant and ε_t is the white noise.

The construction of ARIMA model The flow of the arima model is shown in the figure4

1. Determine the number of differentials d

Initially, we assessed the smoothness of the series and visualized its autocorrelation function in Figure 5(a). Furthermore, we executed a unit root test on it, and both outcomes

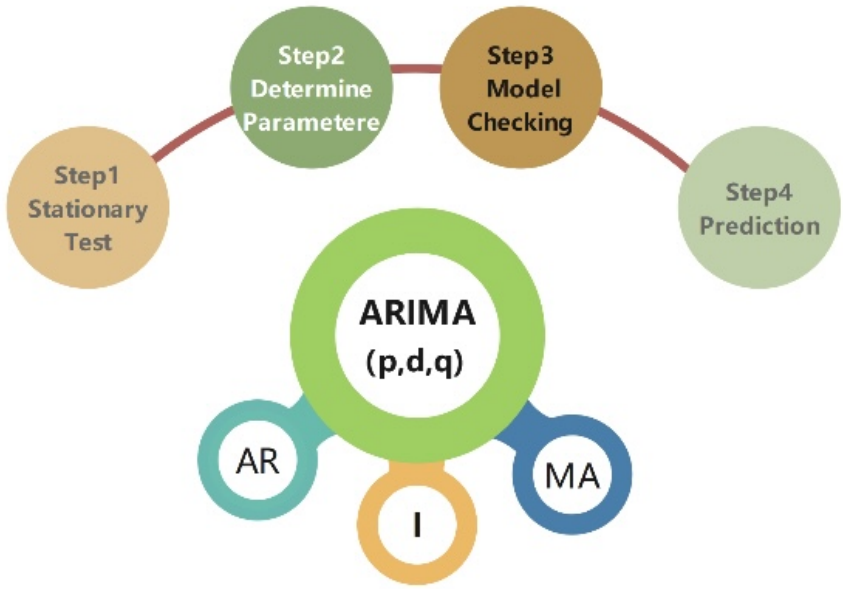
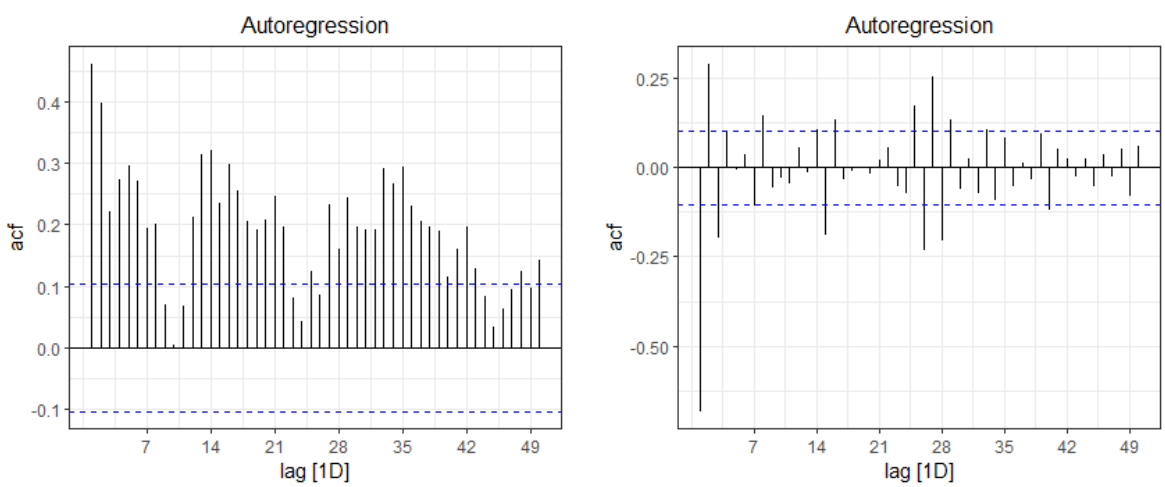


Figure 4: The work flow of ARIMA model

indicated that the series was non-stationary. Therefore, differencing was necessary, and upon executing a first-order differencing, we visualized its autocorrelation function in Figure 5(b). Once again, we conducted a unit root test, and the results indicated that the differenced series attained stationarity. As a result, we set d=1.



(a) ACF before differential

(b) ACF after differential

Figure 5: ACF charts

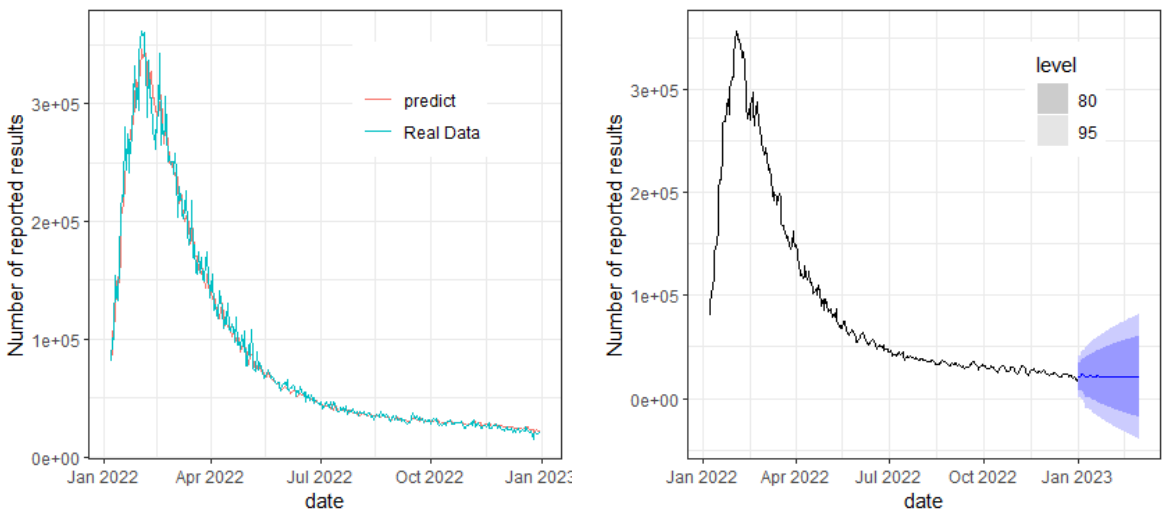
2. Determine the oder of p,q

In statistical modelling, the Akaike information criterion (AIC) is a widely used metric

to evaluate the quality of a model's fit. This criterion is based on the likelihood function and the number of parameters employed in the model. AIC balances model complexity and goodness of fit, with a lower AIC value indicating a model that more accurately fits the data. We set a series of values of p, q. Using a computer program, we fitted these models and subsequently computed their AIC values. The final selection was made by identifying the parameters with the lowest AIC value.After establishing the optimal parameters, a test t-test was conducted to assess their significance, and subsequently, the model was utilized to compute the fitted value of y_t on day t which is denoted as $\hat{y}_t$

2.3 Combined SEIRS-ARIMA model

Upon constructing the SEIRS model and ARIMA model, their association facilitates achieving a more efficient fit to x_t by equation $\hat{x}_t = \hat{y}_t + I_t$.The outcomes of the model fitting are displayed in Figure 6(a), which reveals that incorporating the ARIMA model enhances the SEIRS model's fitting, demonstrating our proposed model's validity.Additionally, Figure 6(b) shows the prediction effect and its confidence interval.For date March 1, 2023 our prediction is 20880.5,and the cofidence interval is (-18757,60518).We have a negative number, which seems incredible, but the main reason is that the variance increases because the forecast time is too long.



(a) Fitting results (b) Predictions and confidence intervals

Figure 6: The result of SEIRS-ARIMA model

3 The attributes affects the percentage

To find the attributes of the word affect the percentage of scores reported that were played in Hard Mode,we extracted some features of the words from given d ata, including frequency, word nature, and letter distribution.From Figure 7,we find the frequency of use of different letters is different, based on this, we divide the letters into frequently used letters (the first four) and infrequently used letters (the last four). We then counted the number of occurrences of high-frequency letters versus low-frequency letters for each

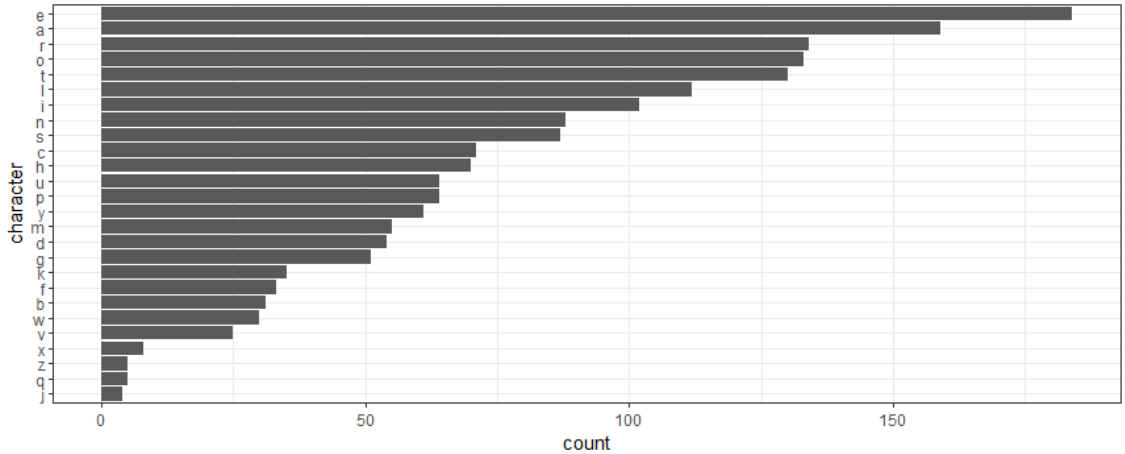
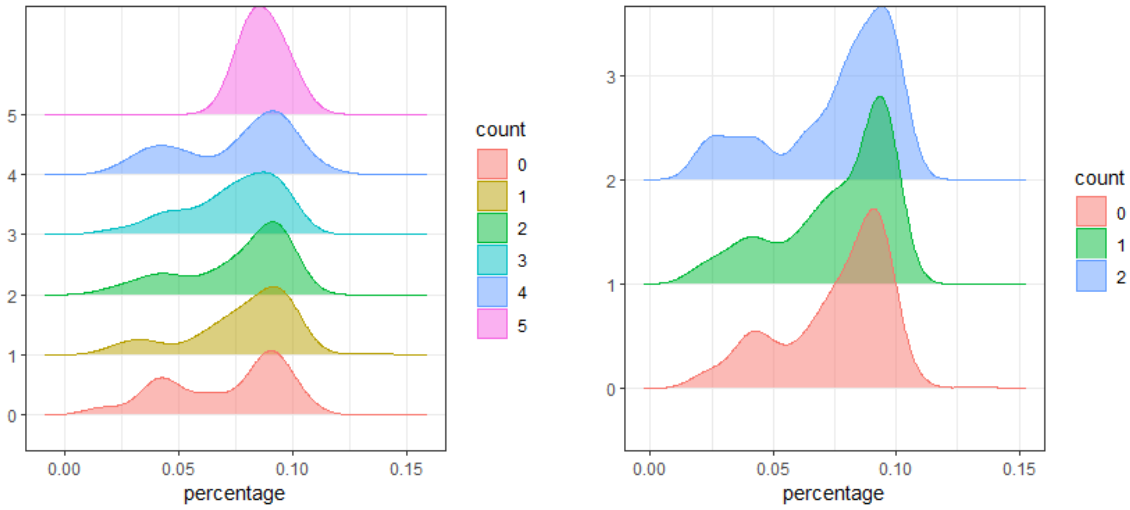


Figure 7: The count of different character in different position

word and drew the distribution of the percentage of scores reported that were played in Hard Mode.Figure 2 shows this percentage in containing different numbers of high-frequency letters,while Figure shows this percentage in containing different numbers of low-frequency letters.

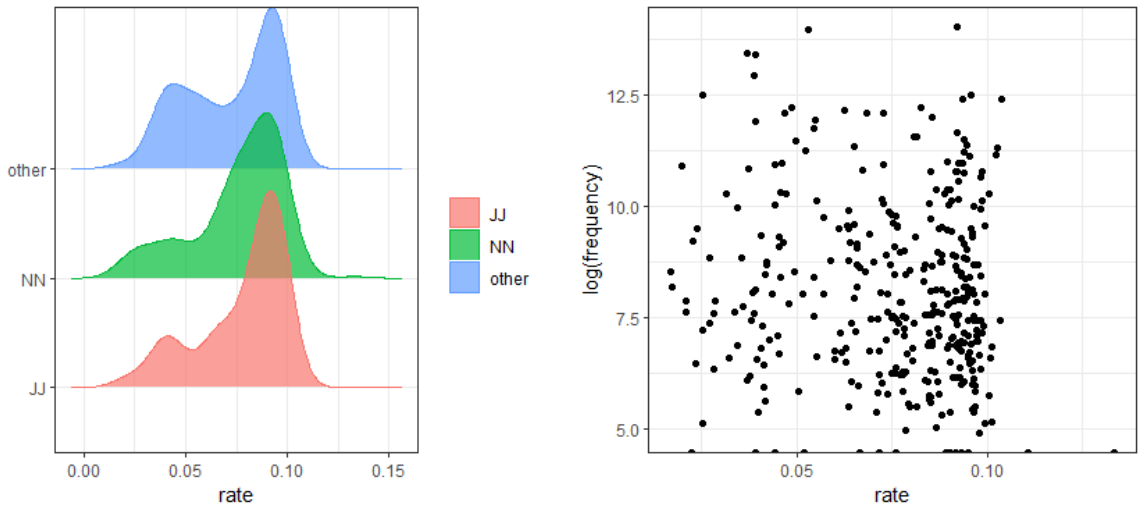


(a) The distribution in different high frequency (b) The distribution in different low frequency

Figure 8: the percentage of scores reported that were played in Hard Mode in different circumstance

From the figure we can see that they do not seem to have any great correlation, we performed an ANOVA on them and the pvalue of the results are all greater than 0.05, which means that we cannot reject the original hypothesis and they are unrelated.For the feature of lexicality, we use the same method, and its density map is shown in the Figure 9(a),The results of the ANOVA likewise tended not to reject the original hypothesis.Finally we discuss the relationship between this percentile and word frequency, and their scatter

plots are shown in the figure 9(b) (due to the excessive word frequency of some words, we performed a logarithmic transformation).We also performed a correlation test (using the Spearman correlation coefficient) and the results remained uncorrelated.Therefore we believe that the attributes of the word are not the main reason that affects it



(a) The distribution in different lexicality

(b) The relationship between percentage and frequency

Figure 9: The relationship of percentage and different attributes

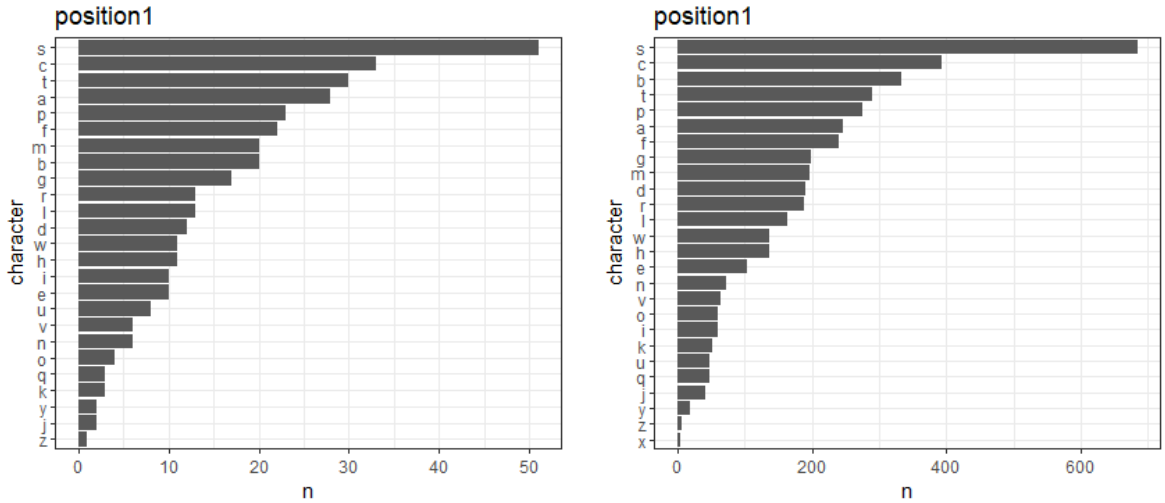
4 Prediction of "tries" of words

4.1 Feature engineering

Problem2 necessitates the anticipation of the distribution of reported results. To this end, we formulate it as a regression problem, and regression analysis is implemented to predict the distribution of frequency percentage associated with all players' attempts. The outcome sought is the prediction of the probability distribution of the number of attempts that an average player requires to play, thereby providing an assessment of the probability of successful attempts.Obviously,the number of attempts reflect the difficulty of this word.We explore four key features which describe the difficulty of this word.

First we calculated the average number of attempts per word $\bar{T}$.This reflects the difficulty of a word in the wordle game.Then we look for features associated with it, which will play a crucial role in the prediction of the model.We found that the letter frequencies at each position in the dataset had a similar distribution to the letter frequencies at the same position in the corpus.This can be seen in Figure 10(here we only show the position1,other positions have the similar attributes).Based on this feature, we define frequency-based scores for different letters in different positions, and they are the open squares of the frequencies of each letter in each position in the corpus.

In addition, considering that if there are many words similar to a certain answer, and



(a) The frequency of characters in position1 from given data

(b) The frequency of characters in position1 from corpus

Figure 10: The similarity between data and corpus in character

these words also have a relatively high frequency, then such words are considered easy to make mistakes. So, we first defined similar words, which have the same letters in a certain three positions, as shown in the figure 11.

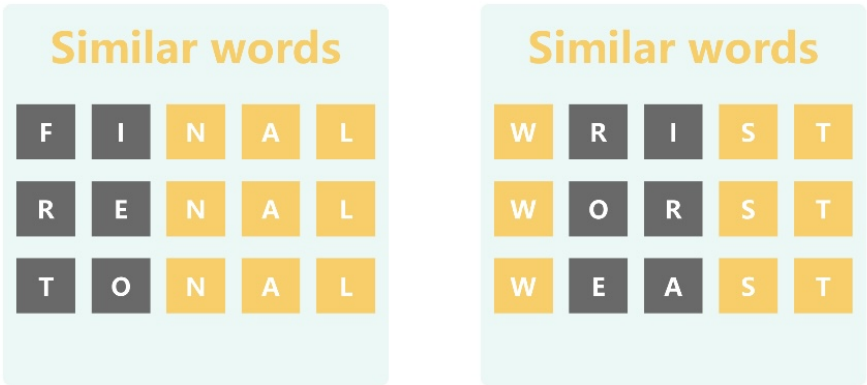


Figure 11: Some example of similar word

We used Lasso regression, hypothesis testing and correlation analysis to screen the potential characteristics proposed above, and performed linear regressions on the screened independent variables to further elaborate how these independent variables affect the distribution of the dependent variable.

Ultimately, we identified four features, all of which have a significant impact on the mean square error produced by the model on the validation set, for a total of nine independent variables: the logarithmic value of the word frequency in the corpus, the lexical-

ity of the word, the letter frequency score in each position of the word and the similarity score of the word. Among them, since most of the words in the dataset were judged as adjectives or nouns by NLTK, we used two variables that took values of only 0 or 1 to indicate whether the words were nouns or adjectives, and if both variables were 0 then the words were considered neither adjectives nor nouns and were classified as other categories.

It can be seen that letting the maximum depth of the tree 7 should be an exciting choice. After similar treatment for the other hyperparameters, we finally determined that the decision tree has the best predictive power when setting hyperparameters as follow.

$\hat{\beta}$ / y	log(Frequency)	is it NN?	is it JJ?	char_freq 1	char_freq 2	char_freq 3	char_freq 4	char_freq 5	similarity score
1 tris	0.05947	0.04317	0.07420	0.01236	0.00171	0.00034	0.00199	0.00276	0.24387
2 tris	0.14400	-0.25801	0.52735	0.11275	0.18974	0.03941	0.10137	0.08103	0.14102
3 tris	0.24667	0.93007	0.73535	0.21011	0.21566	0.06712	0.23705	0.27297	0.42075
4 tris	0.65105	1.23727	-0.12100	-0.03258	-0.12496	-0.02978	0.05437	0.04647	-0.52393
5 tris	-0.19367	0.10752	-0.55166	-0.17905	-0.17699	-0.06220	-0.19021	-0.18333	-0.25973
6 tris	-1.19112	-1.50624	-0.42196	-0.11115	-0.15120	-0.01668	-0.16729	-0.19706	0.26767
7 or more tries	-0.29802	-1.08619	-0.27587	-0.01918	0.00548	0.01567	-0.02466	-0.00469	-0.16190

Table 1: Coefficients of the Linear Regression $\boldsymbol{y} \sim \hat{\boldsymbol{\beta}}\boldsymbol{X}$

4.2 Comparison of multiple regression models

To further explain the rationality of the selected features, we take linear regression as an example to analyze its regression coefficients. From a symbolic point of view, it is obvious that the frequency of "tries" has a significant impact on the distribution of the dependent variable: for the independent variables corresponding to the first four dependent variables, the regression coefficients corresponding to the frequency of "tries" are all positive, while for the independent variables corresponding to the last three dependent variables, the regression coefficients corresponding to the frequency of "tries" are all negative. This indicates that on average, the more frequently a word appears in English, the easier it is for people to think of it and guess it in the early steps of the game. Conversely, less frequent words are less likely to come to mind, resulting in an increase in the number of "tries" needed to guess the word. Similar characteristics can also be observed for word types, with nouns and adjectives being easier to guess in the early steps. The same applies to letter frequency scores. These findings are consistent with our understanding and are reasonable.

For the data features we extracted above, and considering the characteristics of the small data set size, after fully comparing various regression algorithms, we finally chose decision tree regression.

After preliminary and exploratory tries of various regression models, it can be seen that the decision tree regression is significantly more promising than other models under the evaluation based on the MSE and KL divergence on the validation set. Note that the model is still being tested preliminary at this stage, so some models with fewer hyperparameters are optimized, but models with more hyperparameters do not specifically determine the value of the parameters, such as the decision tree. What should be

emphasized is that since the decision tree has a lot of hyperparameters, we just set his hyperparameters arbitrarily now, even so it had a good performance on the training set.

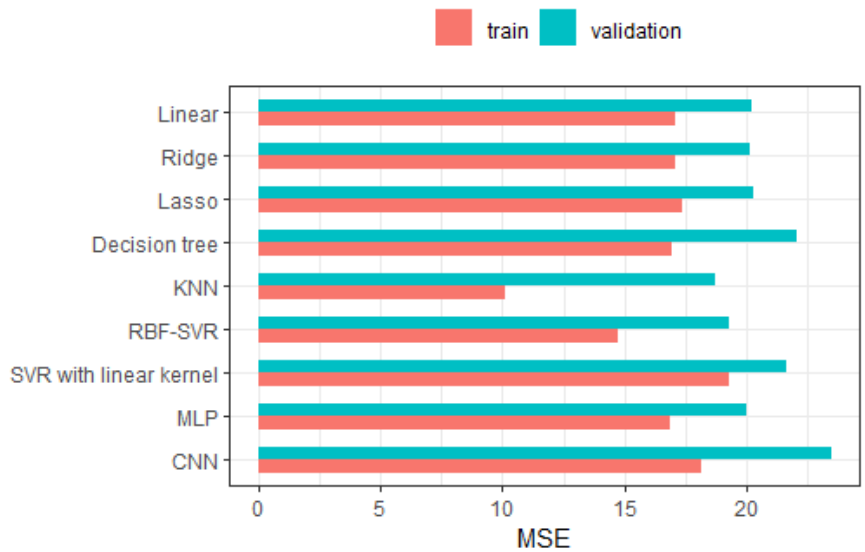


Figure 12: The MSE of different model

Unless otherwise specified, the following model inputs are the features extracted above. Among the models, the hyperparameters of each model are:

1. Linear regression:no hyperparameters to adjust;

2. Ridge regression (l^2 regularized linear regression): use the automatically optimized ridge regression model "RidgeCV" by the library of 'sklearn' in Python, and automatically adjust the hyperparameters;

3. Lasso regression (l^1 regularized linear regression): use the automatically optimized Lasso regression model "LassoCV" by the library of 'sklearn' in Python, and automatically adjust the hyperparameters;

4. Decision tree regression: $max_depth = 3$, $random_state = 0$, other hyperparameters are set by default, so all hyperparameters are not optimized;

5. KNN regression: $K = 4$,the unique hyperparameter 'k' has been adjusted to the optimal value;

6. RBF-SVR: $C = 10.21$, 'C' has been adjusted to the optimal value, and 'sklearn' will adjust automatically the others;

7. SVR with linear kernel: $C = 3.31$, it has been adjusted to the optimal value;

8. MLP, multilayer perceptron: hyperparameters has been adjusted to the optimal values as much as possible;

9. CNN, convolutional neural network: take the words One-Hot encoding as input,

hyperparameters has been adjusted to the optimal values as much as possible.

It is not difficult to see that decision tree regression may have great potential[2].

4.3 Hyperparameters of Tree

The operation of adjusting the hyperparameters that affect the branching of a decision tree is collectively called "pruning". It is easy to know from the theoretical introduction of decision trees that they are prone to serious problems of overfitting, in other words, "there are too many branches", so a good pruning can greatly improve the prediction performance of a decision tree, which is the key to good or bad decision tree performance. In this subsection we will optimize the decision tree to make the most accurate predictions in the above model.

Next we optimize multiple hyperparameters of the decision tree, the most crucial of which is to determine the maximum depth of the tree. By visualizing the relationship between tree depth and MSE, KL divergence, we will choose the best value of the maximum depth of the decision tree through the Figure 13.

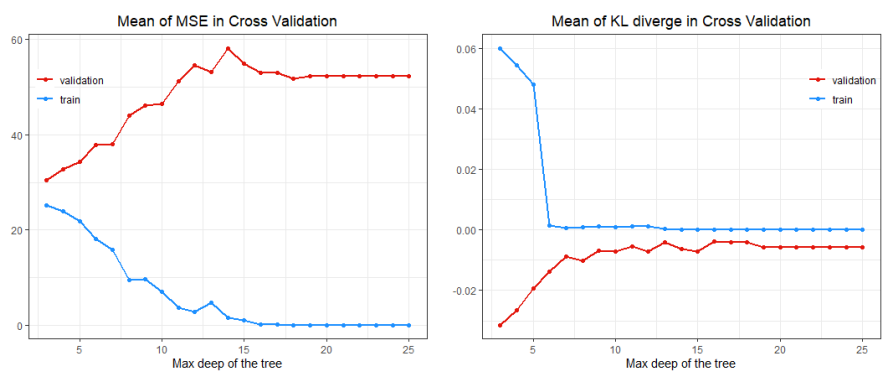


Figure 13: MSE, KL diverge & depth

It can be seen that letting the maximum depth of the tree 7 should be an exciting choice. After similar treatment for the other hyperparameters, we finally determined that the decision tree has the best predictive power when setting hyperparameters as follow.

1.$criterion =' squared_error'$: Since we choose MSE and KL divergence as the evaluation criteria, we choose this hyperparameter as "mean squared error";

2.$max_depth = 7$: Restrict the maximum depth of the tree to 7, which can alleviate overfitting and improve model generalization to a certain extent;

3.$splitter =' random'$: The splits are chosen randomly (choose the best random split every time), which is an important hyperparameter that we can get far better than other models through decision tree regression on such a Small Sample Size. By choosing the splits randomly to some extent, it is possible for the model to learn some potential and hidden data features, which greatly alleviates the overfitting situation and improves the model's prediction ability for data other than training data[1];

4.$min_samples_leaf = 3$: Limit each internal node to contain at least 3 samples, otherwise there will be no branching, which is a means to deal with the problem of overfitting caused by small data set size;

In order to illustrate that our decision tree does have the best regression ability on the above hyperparameters, we compare our new decision tree regression model with the KNN regression and RBF-SVR that performed relatively well:

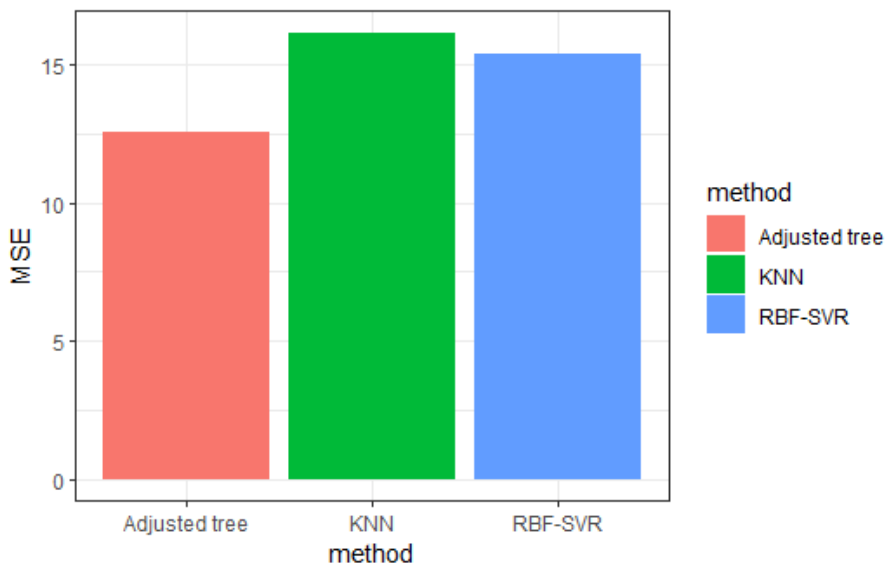


Figure 14: Model comparison

Randomly select 2 data from the validation set as examples for comparison:

begintable

Values / Model	Sample_1						
The true values	0	6	28	37	21	7	1
CART predictions	0.486	7.270	26.567	34.946	21.459	8.027	1.162
SVR predictions	0.118	3.997	19.814	34.263	23.847	9.664	1.544
KNN predictions	0.465	5.687	22.051	32.798	24.020	11.687	3.172
Values / Model	Sample_2						
The true values	6	17	27	27	18	5	1
CART prediction	7.138	21.351	29.401	24.290	15.034	3.127	0.667
SVR predictions	5.670	21.619	33.244	24.900	11.766	1.347	0.000
KNN predictions	6.000	22.202	32.626	24.010	11.677	2.818	0.535

As we can see, the adjusted decision tree has relatively ideal performance, which is why we choose decision tree regression.

4.4 Prediction of "errie"

Finally, let's use our decision tree to make a prediction for the word "errie":

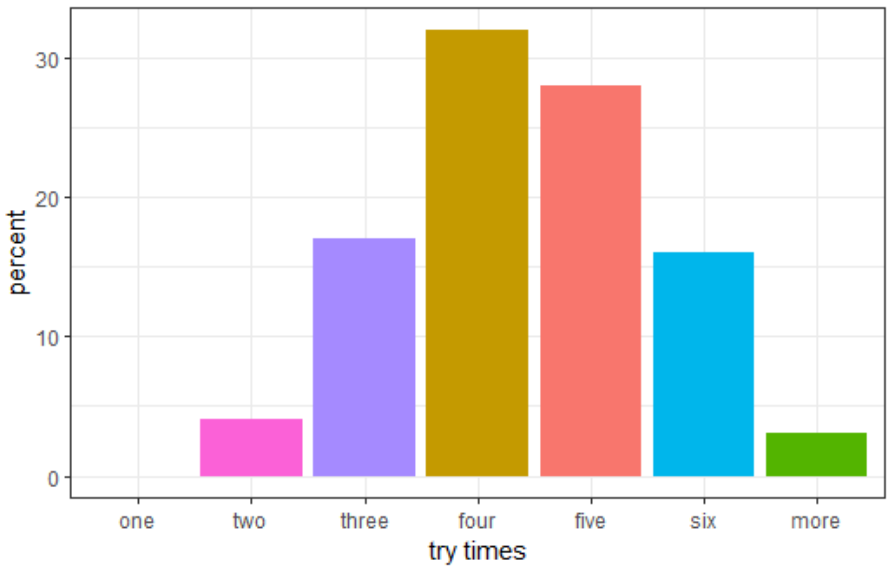


Figure 15: The distribution of each group

The result of prediction is: "1 tries" for 0, "2 tries" for 4, "3 tries" for 17, "4 tries" for 32, "5 tries" for 28, "6 tries" for 16, and "7 tries or more" for 3.

We have a certain degree of certainty in the predicted results, but there will be some deviations from the foreseeable real results, which is mainly caused by the small data set size. However, compared with other models, we are very confident that our model will get more accurate results than them. After all, our model is the best performing one among several regression models. The problem of overfitting is inevitable, but we chose the way that can best alleviate the problem of overfitting[2][1].

In the previous comparison, we can summarize the following advantages and shortcomings of our decision tree regression model:

· Advantages

1. Compared to other models, the MSE on the training set is small enough to suggest that it has better predictive ability than other models.

2. The structure is relatively simple and highly scalable, for example, by adding more branches or further pruning.

3. It is more adaptable to small sample sizes. By adjusting hyperparameters, overfitting can be alleviated without significantly reducing the quality of predictions. Furthermore, the randomness of branching makes it more likely for the model to produce decision trees that can predict the dependent variable well compared to other models.

· Areas for improvement and shortcomings

1. More features can be extracted as input for the decision tree.

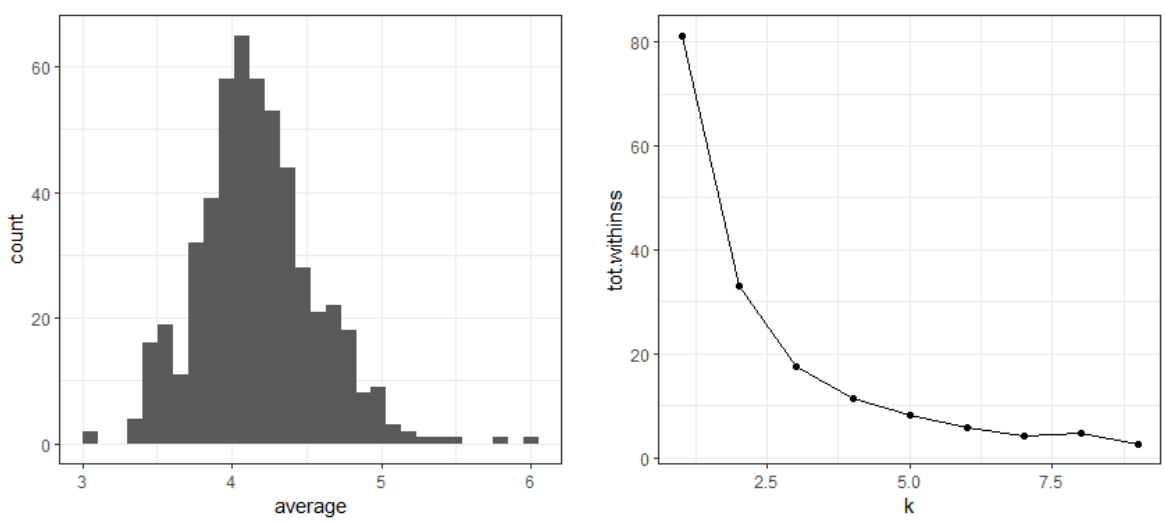
2. The model's interpretability is relatively basic and lacks the intuitiveness of a coefficient interpretation in a generalized linear model.

5 Difficulty differentiation model for words

Determining the level of difficulty of a word is a issue that warrants further investigation. In the this section we develop and summarize a model to classify solution words by difficulty.

5.1 K-means to determine the number of classifications

A challenge in the classification of words by difficulty is the accurate determination of their difficulty level. A key variable that is known to correlate with the level of difficulty is the percentage of guesses from given data. Through the analysis of the percentage, we are able to derive the mean frequency of guesses required for each word,the distribution depicted in Figure 16(a).we employed the k-means method to partition the mean number of required guesses into distinct difficulty clusters. Given that the K-means algorithm is prone to being trapped in a local optima due to its random initial point selection[7], we conducted multiple computations to ensure a globally optimal solution.The results consistently revealed that a three-category partition produced the optimal results.Figure 16(b)shows one of the Scree plots



(a) The distribution of average try

(b) Scree plots from k-means

Figure 16: Distribution and Scree plots

5.2 Gaussian mixture model

Although k-means has been able to provide a difficulty category, this method yield a relatively coarse categorisation and overlook other features available about the words.

To address this shortcoming, we utilised the number of classifications obtained from the k-means algorithm to a Gaussian mixture model, which uses various word features extracted from prior analyses as input. This approach allowed for a more precise and detailed classification of the words.

A Gaussian mixture model can be considered as a model that is a combination of K single Gaussian models, and these K sub-models are the hidden variable of the mixture model.The probability density function of GMM is as follows[6]:

$$p(x) = \sum_{i=1}^{K} p(i)p(x \mid i) = \sum_{i=1}^{K} \pi_i N\left(x, u_i, d_i\right)$$

where

- x_j denotes the jth observation
- K is the number of sub-Gaussian models in the mixture model
- a_k is the probability that the observation belongs to the kth submodel
- $\phi\left(x \mid \theta_k\right)$is the probability density of the kth submodel,θ_k is its mean and variance
- γ_{jk} means that the jth observation belongs to the k submodel with probability.These are Laten variables

Given the existence of latent variables, we considered the use of the Expectation-Maximization (EM) algorithm to address the classification problem. We use a computer program for iterative solving.The results show that the last three groups have 66, 33, and 220 words respectively, and their distribution can be seen in Figure 17 Based on the re-

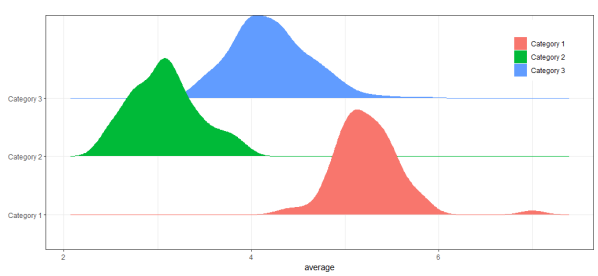


Figure 17: The distribution of each group

sult, it is evident that the cohort categorized as group 1 demonstrates the highest average number of attempts, a factor that we infer to indicate its elevated level of difficulty. Conversely, group 2 displays the lowest average number of attempts, suggesting its status as a less challenging group. Group 3 falls in the intermediate position leading us to categorize it as a moderately difficult group.Randomly extract two words from each of the three categories to see their features:

Groups \ Attribute	Wrods	1tris	2 tris	3 tris	4 tris	5tris	6 tris	7 or more tries
Group 1	gamer	1	4	17	28	26	18	6
	night	1	8	27	27	17	13	7
Group 2	their	2	14	36	30	13	4	0
	point	1	16	37	28	12	4	1
Group 3	crimp	1	5	28	38	20	7	1
	sugar	1	6	25	34	23	9	1

Table 2: Examples of the Clustering Results

this reflects, to some extent, the validity of our model.Finally, for word **eerie**, which we bring into the model, its grouping shows up as **group 3 (medium group)**, mainly because it is less frequently used with a frequency of 2272, but words similar to it have a lower score, and the letters in each position are those that frequently occur in that position

6 Other features in the data

6.1 Feature 1

While we've explored quite a few features of wordle this game above, the nature of this game makes it have other interesting features worth exploring.In Figure 18 , we plot the percentage change over time in the scores reported in the difficult mode. The graphical

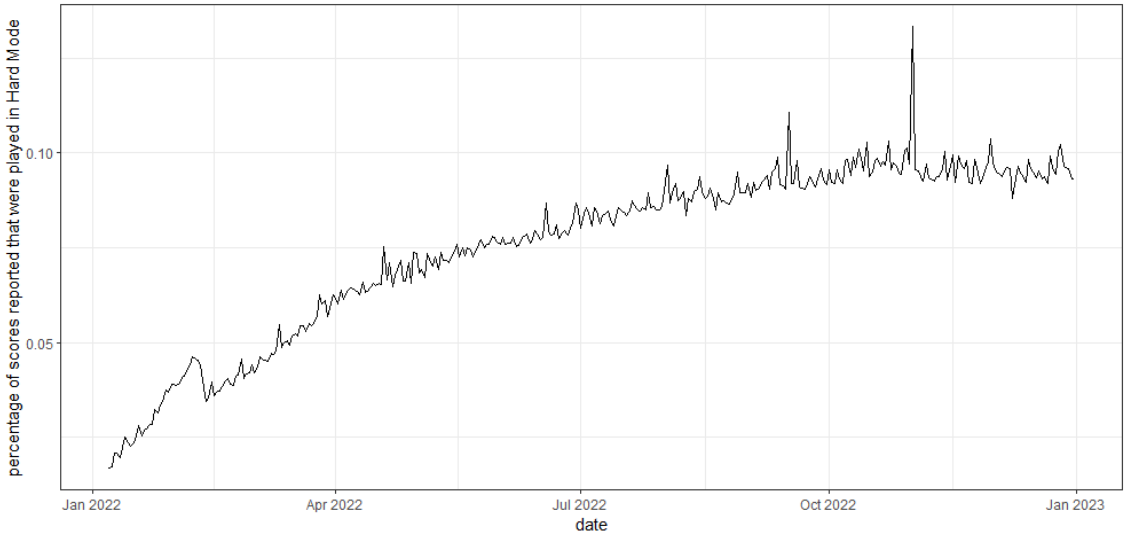


Figure 18: Percentage change over time in the scores reported in the difficult mode

representation illustrates an increasing trend of the percentage change over time in the scores reported in the difficult modeover time. Conversely, 3(a) depicts a decrease in the total number of players in later stages. Notably, the rate of playing hard mode exhibits an upward trend during this period. This phenomenon can be attributed to the persistent

enthusiasm of a subset of dedicated players who maintain their interest in the game even as novelty wanes. Consequently, the total number of players decreases while the retention of core players remains high. The player attrition is primarily due to casual gamers or individuals seeking a brief experience of the game.

6.2 Feature 2

In addition, we plot the variation of the percentage of different attempts over time in Figure It can be seen that the percentage of different attempts shows different fluctuations

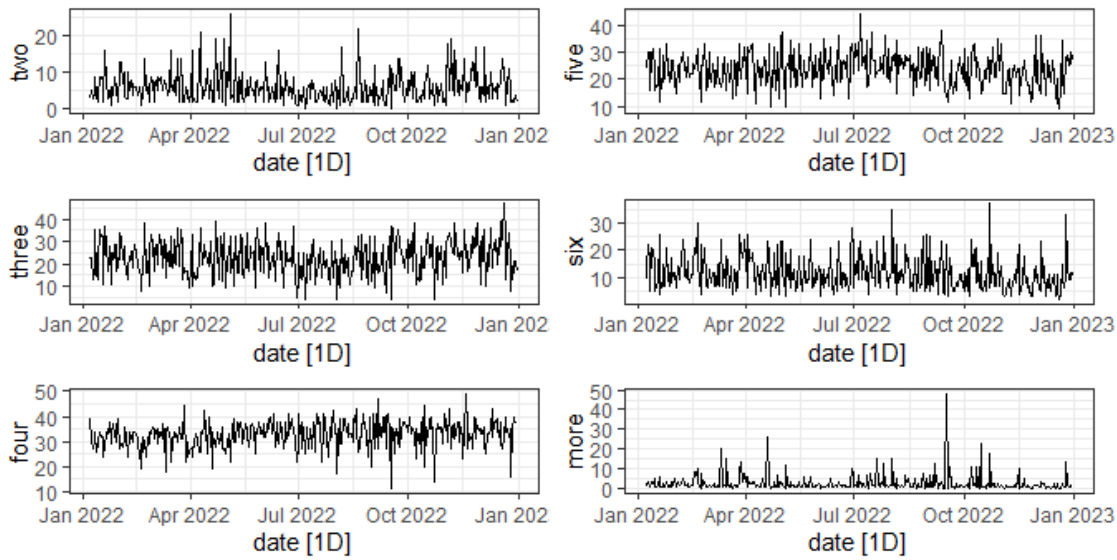


Figure 19: variation of the percentage of different tries

as time moves. Despite the fluctuations, they all try to fluctuate between a standard value, which shows the efforts made by the game creators for the balance of the game.Further, we plotted the average number of attempts over time in Figure 20, which also verifies our judgment that the game has a good balance

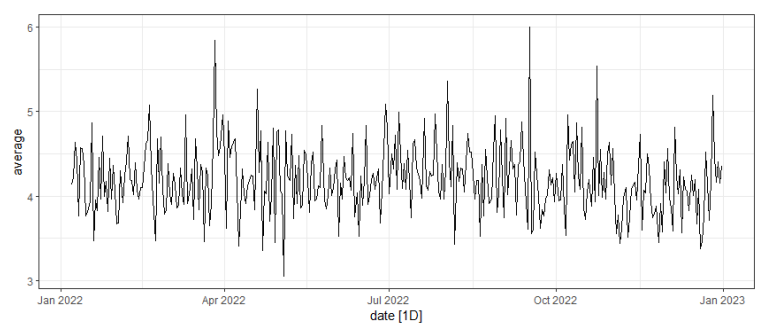


Figure 20: average number of attempts over time

Dear editor:

We are a team of students participating in the 2023 MCM competition. First of all, we would like to say that Wordle is a game that is both fun and challenging. As competitors, we analyzed a series of data sets provided by the organizing committee as well as the COCA corpus, and discovered some clues and information that you might find interesting.

The following is a report and summary of our work. We apologize for any inconvenience.

First, we explored the relationship between the number of users and time. We analyzed the relationship between the number of users who posted their Wordle scores on Twitter each day and the date. Initially, we found that the number of users who posted their Wordle scores on Twitter each day should be proportional to the popularity of the game, indicating that the number of users who posted their Wordle scores on Twitter each day quantified the popularity of the game to some extent.

At the same time, we noticed that the number of people who posted their Wordle scores on Twitter each day had a sharp increase followed by a slow decrease, and then it leveled off. This trend suggests that Wordle rapidly became popular on the internet for a period of time, similar to viral trends, and was then slowly phased out. Considering the similarities between information dissemination and virus transmission, we pioneered the introduction of the SEIRS (Susceptible-Exposed-Infected-Recovered-Susceptible) model from epidemiology, combined with the classic time-series model SARIMA, to explain the sharp increase. A new thing entering the public's view can always quickly attract attention in the initial stage because "fresh information" will be quickly spread and disseminated among the population. The subsequent gradual decrease in the number of players is due to the fact that the public's attention is attracted by newer things, newer "fresh information," and thus the popularity of the game slowly declines.

One way to increase popularity is to focus on promotion, which in our model is to increase the transmission and re-infection rates. Another way is to introduce new gameplay to keep players interested, which in our model is explained as the recovery rate. However, we noticed that although the total number of players decreased after reaching its peak, the proportion of players playing the hard mode increased over time. This is because there are always some players who genuinely like this game, and they will not give up on this game just because the novelty wears off quickly. The total number of players may decrease, but the retention rate of core players after the novelty wears off is high, and most of the players who leave are casual players or those who just want to try it out.

Next, we analyzed the relationship between word characteristics and word guessing accuracy. We explored the distribution of frequency and guessing times and found that the average number of guesses is positively correlated with frequency, which was our first feature. We then analyzed the distribution of word parts of speech and discovered some special rules: the frequency of "s," "t," and "c" in the first position is relatively high, and the frequency of "o," "a," "l," and "r" in the second position is relatively high. The third, fourth, and fifth positions of the letters also have their own special frequency dis-

tributions, which we will not elaborate on here. Based on this, we used the COCA corpus to extract all five-letter words and analyzed their letter distribution in each position, and found that the words in the corpus had the same characteristics. Lastly, we analyzed the relationship between the number of guesses and the length of the word. By analyzing the data provided by the competition committee, we found that the average number of guesses required to solve a word increases with the length of the word, which is consistent with intuition. However, we also found that there is a significant variation in the number of guesses required to solve a word of the same length, which suggests that the difficulty of the word may also depend on its specific combination of letters.

In conclusion, our analysis sheds light on various interesting aspects of the Wordle game, including its popularity dynamics, the relationship between word characteristics and guessing accuracy, and the variation in the difficulty of the words. We believe that these findings could be useful for future studies of language and cognition, as well as for improving the design and marketing of similar games. Thank you for your attention and consideration.

Sincerely,

A team from MCM competitors.

References

[1] Myles, Anthony J., Robert N. Feudale, Yang Liu, Nathaniel A. Woody, and Steven D. Brown. "An introduction to decision tree modeling." Journal of Chemometrics: A Journal of the Chemometrics Society 18, no. 6 (2004): 275-285.

[2] Khemphila, A., & Boonjing, V. (2010, October). Comparing performances of logistic regression, decision trees, and neural networks for classifying heart disease patients. In 2010 international conference on computer information systems and industrial management applications (CISIM) (pp. 193-198). IEEE.

[3] Wang, W. (2002). Global behavior of an SEIRS epidemic model with time delays. Applied Mathematics Letters, 15(4), 423-428.

[4] Xu, R., & Ma, Z. (2010). Global stability of a delayed SEIRS epidemic model with saturation incidence rate. Nonlinear Dynamics, 61(1-2), 229-239.

[5] Ho, S. L., & Xie, M. (1998). The use of ARIMA models for reliability forecasting and analysis. Computers & industrial engineering, 35(1-2), 213-216.

[6] Baum, C. F., Schaffer, M. E., & Stillman, S. (2003). Instrumental variables and GMM: Estimation and testing. The Stata Journal, 3(1), 1-31.

[7] Xu, J., & Liu, H. (2010, October). Web user clustering analysis based on KMeans algorithm. In 2010 International Conference on Information, Networking and Automation (ICINA).

[8] Wei, D., Chen, F., & Lin, L. (2018). Simulation of Hot Event Propagation Based on Game Theory and SIRS. Journal of System Simulation, 30(6), 2050.

A Search for the best value of Decision Tree Depth

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_squared_error
from scipy.stats import entropy
from sklearn.model_selection import cross_val_score
from sklearn.metrics import make_scorer

def kl_divergence(y_true, y_pred):
hist_true, _ = np.histogram(y_true, bins=2, density=True)
hist_pred, _ = np.histogram(y_pred, bins=2, density=True)
return entropy(hist_true, hist_pred)

X_total = words_freq.iloc[:, 3:].values
y_total = np.array([word for word in words_freq['Proportion']])

K = 26
heights = np.arange(3, K, 1)
```

```
mse_scores = []
kl_scores = []

for i in heights:
testTree = DecisionTreeRegressor(
max_depth=i,
criterion='squared_error',
splitter='random',
random_state=0
)
testTree.fit(X_total, y_total)
mse = mean_squared_error(y_total, testTree.predict(X_total))
mse_cv = np.mean(cross_val_score(
testTree,
X_total,
y_total,
cv=3,
scoring=make_scorer(mean_squared_error)
))
kl = kl_divergence(y_total, testTree.predict(X_total))
kl_cv = np.mean(cross_val_score(
testTree,
X_total,
y_total,
cv=3,
scoring=make_scorer(kl_divergence, greater_is_better=False)
))
mse_scores.append([i, mse, mse_cv])
kl_scores.append([i, kl, kl_cv])

mse_scores = np.array(mse_scores)
kl_scores = np.array(kl_scores)
```